AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

MT101
General Mathematics

Midterm Examination (MTA)
Summer Semester 2025/2026

Time Allowed: 90 minutes

Instructions:

- Total Marks:60
- This exam consists of 4 questions.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Calculators** are allowed.

AOU EGYPT

الجامعة العربية المفتوحة - مصر
Arab Open University
القبول والتسجيل
الامتحانات
Arab Republic of Egypt

CamScanner

ESSAY QUESTIONS

Each question is worth 15 marks. You should answer all the questions, giving 60 possible marks.

Q–1: [4+5+6 Marks]

a) perform the indicated operations.

i) Let $f(x)= x + \frac{1}{x}$ find $f(1+3i)$ (express the result as a+ bi)

ii) simplify: $3(-2+i) - 4(3-2i)$. (express the result as a+ bi)

b) Solve for x: $2(5x-3) - x = 3(3x-2)$

Q–2: [7+8Marks] solve the following:

a) $|9 + 3x| < 4$ (write the solution in interval notation)

b) $0 < 1 - x \leq 3$ or $-1 \leq 2x - 3 \leq 5$(write the solution in interval notation)

Q–3: [7+8] Marks] a) solve the following equations:

a) $\sqrt[3]{3x+1} - 1 = 4$

b) $x^2 - 2x - 5 = 0$ (use completing the square)

Q–4: [8+7Marks]

a) the point (1, 4) is on a circle whose center is (0,0), write the equation of the circle. .

b) Find the equation of the line with x-intercept 1 and y-intercept -3.

End of the questions.

AOU EGYPT

الجامعة العربية المفتوحة - مصر
Arab Open University
القبول والتسجيل
الامتحانات
Arab Republic of Egypt

CamScanner